Oxford Nanopore Technologies

PROTOCOL

# Native barcoding DNA V14 – automated ElysION (SQK-NBD114.96)

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Automated native barcoding method using the ElysION device, outlining library preparation and sequencing.

This protocol:

- Enables multiplexing of up to 48 samples
- Allows analysis of native DNA
- Is a PCR-free method
- Is compatible with R10.4.1 flow cells

**For Research Use Only**

This method is used with an Early Access device.

For more information about our Early Access programmes, please refer to this article on product release phases.

**FOR RESEARCH USE ONLY**

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# 1. Overview of the protocol

**This protocol is a work in progress and some details are expected to change over time. Please make sure you always use the most recent version of the protocol.**

### Introduction to the automated native barcoding DNA protocol using ElysION

The ElysION automated Native Barcoding Kit 96 V14 (SQK-NBD114.96) workflow is optimised for simplicity and speed to automatically prepare and sequence between 1 and 48 purified DNA input samples, then automatically wash the used flow cell for re-use. The workflow is PCR-free, removing the PCR bias and retains information about base modifications, which can be analysed using tools developed in the Nanopore Community.

**Note:** This kit has been optimised to generate long reads and high sequencing accuracies of over 99% (Q20+). However, in order for long reads to be observed in sequencing, long fragments need to be present in the initial sample input. Alternatively, if users require a faster speed to sequencing, we recommend using our Rapid Barcoding Kit 96 V14 kit (SQK-RBK114.96).

## Steps in the sequencing workflow:

### Prepare for your experiment

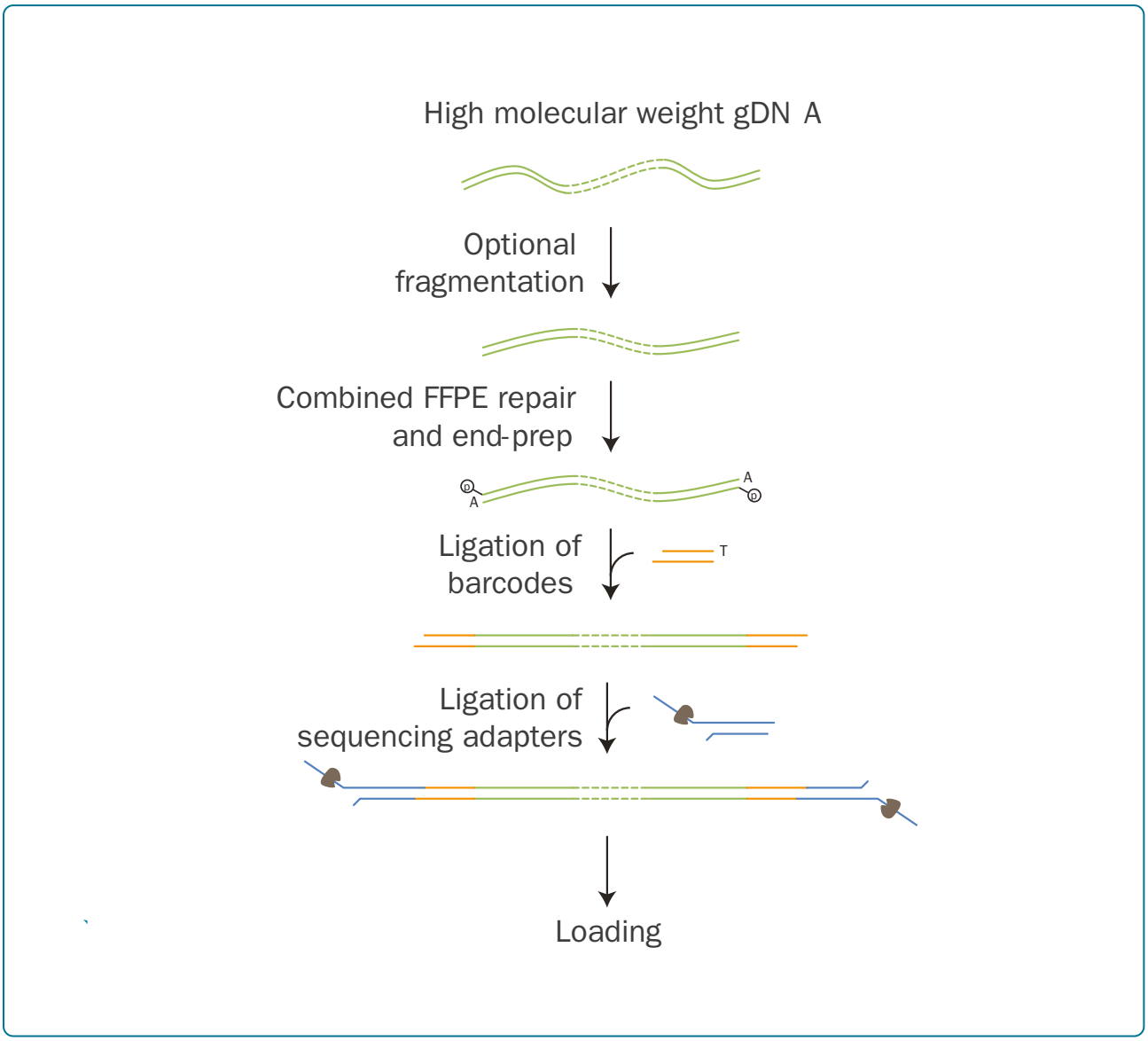
You will need to:

- Ensure your ElysION device is installed with the required hardware and software package for this workflow.
- Ensure you have your sequencing kit, the correct equipment and third-party reagents.
- Check your flow cell to ensure it has enough pores for a good sequencing run.

### Automated library preparation, sequencing and analysis

The Table below is an overview of the steps automated by the ElysION device.

| Library preparation step | Process |
|---|---|
| DNA repair and end-prep | Reparation the DNA, preparation the DNA ends for adapter attachment, and bead clean-up. |
| Native barcode ligation and clean-up | Attachment of the native barcodes to the DNA ends and bead clean-up. |
| Adapter ligation and clean-up | Attachment of the sequencing adapters to the DNA ends and bead clean-up. |
| Priming and loading the flow cell | Prime the flow cell and load the prepared library for sequencing. |
| Sequencing | Your sequencing run uses the Gourami software, which will collect raw data to basecall and demultiplex the barcoded reads. |
| Data analysis | Once sequencing is complete, we have a number of downstream analysis options for your data, available through EPI2ME. |




**Compatibility of this protocol**

This protocol should only be used in combination with:

- Native Barcoding Kit 96 V14 (SQK-NBD114.96)
- R10.4.1 flow cells (FLO-MIN114)
- Flow Cell Wash Kit (EXP-WSH004)
- Flow Cell Priming Kit V14 (EXP-FLP004)
- Sequencing Auxiliary Vials V14 (EXP-AUX003)
- ElysION device

## 2. Equipment and consumables

**Materials** For 1–4 samples: 270 fmol DNA in 20 µl per sample

| | |
|---|---|
| | For 5–48 samples: 110 fmol DNA in 20 µl per sample<br>Native Barcoding Kit 96 V14 (SQK-NBD114.96)<br>Flow Cell Wash Kit (EXP-WSH004) |
| **Consumables** | MinION/GridION Flow Cell<br>NEB Blunt/TA Ligase Master Mix (NEB, M0367)<br>NEBNext FFPE Repair Mix (NEB, M6630)<br>NEBNext Ultra II End repair/dA-tailing Module (NEB, E7546)<br>NEBNext Quick Ligation Module (NEB, E6056)<br>Hard-Shell® 96-Well PCR Plates, low profile, thin wall, skirted, red/clear (Bio-Rad™, HSP9611)<br>Nunc™ 96 Well Polypropylene 2 ml DeepWell Plate (Thermo Scientific, 95040452)<br>ElysION Disposable Tips - 1000 µl (ELY-TIP1000)<br>ElysION Disposable Tips - 200 µl (ELY-TIP0200)<br>ElysION Disposable Tips - 50 µl (ELY-TIP0050)<br>ElysION Disposable Tips Box Small (ELY-TBS01)<br>ElysION Disposable Tips Box Large (ELY-TBL01)<br>Reservoir Plate (Porvair, 390015)<br>Azenta PCR Plate Lid (Azenta, 4ti-0291)<br>Screw Cap Micro tube 2 ml, sterile (Sarstedt™, 72.694.006)<br>Screw Cap Micro tube 0.5 ml, sterile (Sarstedt™, 72.730.106)<br>ElysION Waste Bin (ELY-WB01)<br>Bovine Serum Albumin (BSA) (50 mg/ml) (e.g Invitrogen™ UltraPure™ BSA 50 mg/ml, AM2616)<br>Freshly prepared 80% ethanol in nuclease-free water<br>Nuclease-free water (e.g. Thermo Scientific, AM9937)<br>1.5 ml Eppendorf DNA LoBind tubes |
| **Equipment** | ElysION device with MinION integration<br>MinION/GridION Flow Cell Light Shield<br>Microfuge<br>Vortex mixer<br>Microplate centrifuge, e.g. Fisherbrand™ Mini Plate Spinner Centrifuge (Fisher Scientific,11766427)<br>Multichannel pipette and tips<br>P1000 pipette and tips<br>P200 pipette and tips<br>P100 pipette and tips<br>P20 pipette and tips<br>P2 pipette and tips<br>Ice bucket with ice |
| **Optional equipment** | Qubit™ fluorometer (or equivalent for QC check) |

**For this protocol you will require the following sample input:**

We strongly recommend loading your samples to a fixed molarity for the best experiment results:

- For 1-4 samples: 270 fmol DNA in 20 µl per sample
- For 5-48 samples: 110 fmol DNA in 20 µl per sample

However, if the molarity of your samples cannot be calculated, please follow the recommendations below. Please note, we recommend using molarity for optimal sample input. The input recommendations below are based on sample input being approximately 10 kb in length.

- For 1-4 samples: 1,666 ng DNA in 20 µl per sample (83.3 ng/µl)
- For 5-48 samples: 333 ng DNA in 20 µl per sample (16.6 ng/µl)

**Note:** The output, and sequencing read lengths of extracted DNA may vary depending on sample quality and species. Please ensure you are using high-quality sample inputs.

#### Sample type method variables: gDNA or amplicon

This method is used for gDNA samples by default. There is an option to select the amplicon sample input variable.

For more information on editing the method configurator variables, please refer to the ElysION User Guide.

Amplicon samples do not require DNA end repair. Therefore, the End-Prep Master Mix formulation will differ depending on your sample type. Please follow the instructions on the on-screen display to ensure you are using the correct method for your run settings.

#### Input DNA

### How to QC your input DNA

It is important that the input DNA meets the quantity and quality requirements. Using too little or too much DNA, or DNA of poor quality (e.g. highly fragmented or containing RNA or chemical contaminants) can affect your library preparation.

For instructions on how to perform quality control of your DNA sample, please read the Input DNA/RNA QC protocol.

### Chemical contaminants

Depending on how the DNA is extracted from the raw sample, certain chemical contaminants may remain in the purified DNA, which can affect library preparation efficiency and sequencing quality. Read more about contaminants on the Contaminants page of the Community.

#### Check your flow cell

The number of pores in your flow cell will be checked by the ElysION device at the start of your assay. MinION/GridION Flow Cells should be checked within 12 weeks of purchasing.

Oxford Nanopore Technologies will replace any unused flow cell with fewer than the number of pores listed in the Table below, when the result is reported within two days of performing the

flow cell check, and when the storage recommendations have been followed. To do the flow cell check, please follow the instructions on the ElysION on-screen display.

| Flow cell | Minimum number of active pores covered by warranty |
|---|---|
| MinION/GridION Flow Cell | 800 |

**Third-party reagents**

We have validated and recommend the use of all the third-party reagents used in this protocol. Alternatives have not been tested by Oxford Nanopore Technologies.

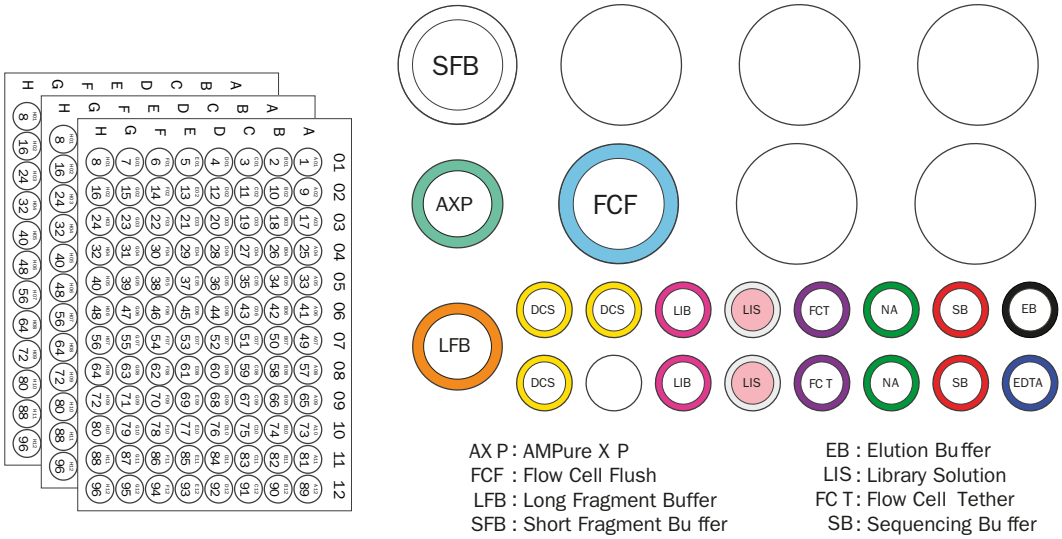
For all third-party reagents, we recommend following the manufacturer's instructions to prepare the reagents for use.

**The Native Adapter (NA) used in this kit and protocol is not interchangeable with other sequencing adapters.**

### Native Barcoding Kit 96 V14 (SQK-NBD114.96) contents

**Note:** We are in the process of updating our native barcoding kits with an increased volume of Short Fragment Buffer (SFB). If you have an old format kit and/or require additional volume of Short Fragment Buffer (SFB), this can be purchased via our **SFB Expansion (EXP-SFB001)**.

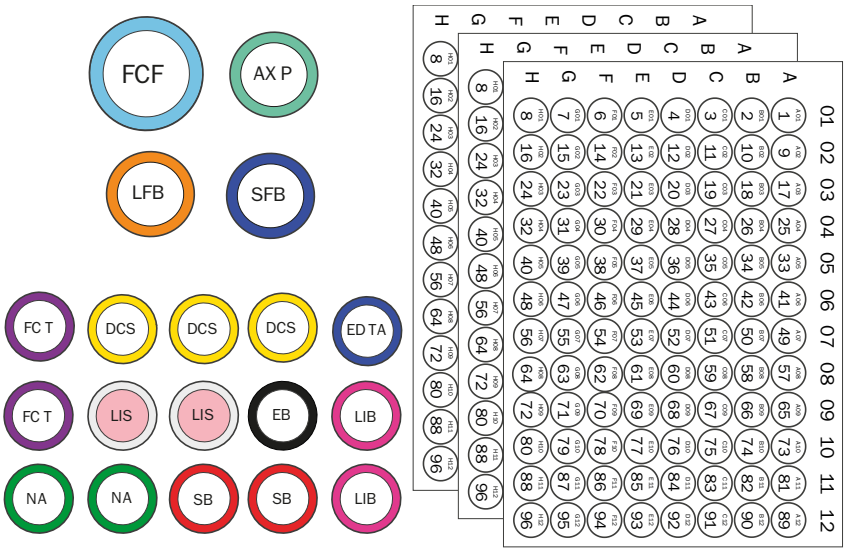
## New format: increased volume of Short Fragment Buffer (SFB)




AXP : AMPure XP
FCF : Flow Cell Flush
LFB : Long Fragment Buffer
SFB : Short Fragment Buffer
EDTA : EDTA
LIB : Library Beads
EB : Elution Buffer
LIS : Library Solution
FCT : Flow Cell Tether
SB : Sequencing Buffer
NA : Native Adapter
DCS : DNA Control Sample

| Name | Acronym | Cap colour | No. of vials | Fill volume per vial (µl) |
|---|---|---|---|---|
| Native Barcode plate | NB01-96 | - | 3 plates | 8 µl per well |
| DNA Control Sample | DCS | Yellow | 3 | 35 |
| Native Adapter | NA | Green | 2 | 40 |
| Sequencing Buffer | SB | Red | 2 | 700 |
| Library Beads | LIB | Pink | 2 | 600 |
| Library Solution | LIS | White cap, pink label | 2 | 600 |
| Elution Buffer | EB | Black | 1 | 1,500 |
| AMPure XP Beads | AXP | Clear cap, light teal label | 1 | 6,000 |
| Long Fragment Buffer | LFB | Clear cap, orange label | 1 | 7,500 |
| Short Fragment Buffer | SFB | Clear | 1 | 25,000 |
| EDTA | EDTA | Blue | 1 | 700 |
| Flow Cell Flush | FCF | Clear cap, light blue label | 1 | 15,500 |
| Flow Cell Tether | FCT | Purple | 2 | 200 |

## Old format: lower volume of Short Fragment Buffer (SFB)




AX P : AMPure X P
FCF : Flow Cell Flush
LFB : Long Fragment Bu ffer
SFB : Short Fragment Bu ffer
ED TA : ED TA
LIB : Library Beads

EB : Elution Bu ffer
LIS : Library Solution
FC T : Flow Cell Tether
SB : Sequencing Bu ffer
NA : Native Adapter
DCS : DNA Control Sample

| Name | Acronym | Cap colour | No. of vials | Fill volume per vial (µl) |
|---|---|---|---|---|
| Native Barcode plate | NB01-96 | - | 3 plates | 8 µl per well |
| DNA Control Sample | DCS | Yellow | 3 | 35 |
| Native Adapter | NA | Green | 2 | 40 |
| Sequencing Buffer | SB | Red | 2 | 700 |
| Library Beads | LIB | Pink | 2 | 600 |
| Library Solution | LIS | White cap, pink label | 2 | 600 |
| Elution Buffer | EB | Black | 1 | 1,500 |
| AMPure XP Beads | AXP | Clear cap, light teal label | 1 | 6,000 |
| Long Fragment Buffer | LFB | Clear cap, orange label | 1 | 7,500 |
| Short Fragment Buffer | SFB | Brown cap, dark blue label | 1 | 7,500 |
| EDTA | EDTA | Blue | 1 | 700 |
| Flow Cell Flush | FCF | Clear cap, light blue label | 1 | 15,500 |
| Flow Cell Tether | FCT | Purple | 2 | 200 |

**Note:** This product contains AMPure XP reagent manufactured by Beckman Coulter, Inc. and can be stored at -20°C with the kit without detriment to reagent stability.

The barcodes are oriented in columns in the barcode plate.

| | 01 | 02 | 03 | 04 | 05 | 06 | 07 | 08 | 09 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A | NB01 | NB09 | NB17 | NB25 | NB33 | NB41 | NB49 | NB57 | NB65 | NB73 | NB81 | NB89 |
| B | NB02 | NB10 | NB18 | NB26 | NB34 | NB42 | NB50 | NB58 | NB66 | NB74 | NB82 | NB90 |
| C | NB03 | NB11 | NB19 | NB27 | NB35 | NB43 | NB51 | NB59 | NB67 | NB75 | NB83 | NB91 |
| D | NB04 | NB12 | NB20 | NB28 | NB36 | NB44 | NB52 | NB60 | NB68 | NB76 | NB84 | NB92 |
| E | NB05 | NB13 | NB21 | NB29 | NB37 | NB45 | NB53 | NB61 | NB69 | NB77 | NB85 | NB93 |
| F | NB06 | NB14 | NB22 | NB30 | NB38 | NB46 | NB54 | NB62 | NB70 | NB78 | NB86 | NB94 |
| G | NB07 | NB15 | NB23 | NB31 | NB39 | NB47 | NB55 | NB63 | NB71 | NB79 | NB87 | NB95 |
| H | NB08 | NB16 | NB24 | NB32 | NB40 | NB48 | NB56 | NB64 | NB72 | NB80 | NB88 | NB96 |

**Note:** The DNA Control Sample (DCS) is a 3.6 kb standard amplicon mapping the 3' end of the Lambda genome.

## 3. ElysION SQK-NBD114.96 sample sheet setup

### Sample sheet information

The sample sheet assigns a barcode to each sample, allowing a user to pick a range from the 96-well plate, and for a sample ID to be tracked from input to results.

For more information on setting up a sample sheet please refer to the ElysION User Guide.

### 1 Set up your sample sheet as a CSV file with the following information:

| Required field | User input | Definition / field info |
|---|---|---|
| **v1** | | Version field, no input required outside of field |
| | | |
| **assay** | nbd:v1.0 | Assay ID and version |
| **library_id** | User-defined | Identification for the DNA library |
| **created_by** | User-defined | Identification of operator setting up sample sheet |
| **created_at** | User-defined | Date and time of creation. Please follow the format outlined in the ElysION user guide. |
| **sample_count** | User-defined | Number of samples processed in run (1 to 48 samples) |
| | | |
| **well_id** | A1-H6 | Positions in 96-well plate being used (1 to 48 samples).<br><br>Samples **must start from position A1** in the 96-well plate and run consecutively column-wise. |
| **barcode** | barcode01-barcode96 | Rapid barcodes from SQK-NBD114.96 being used in library preparation.<br><br>Up to 96 barcodes are available in the sequencing kit, these should be used sequentially. (e.g. Barcodes 01-48, Barcodes 01-24, or Barcodes 49-96) |
| **sample_type** | User-defined | Description or characteristics of sample input |
| **sample_id** | User-defined from LIMS system (lims-sampleid-12345) | Identification for each sample input (e.g. from LIMS system) |

Below is an example of a sample sheet CSV file:

| | A | B | C | D |
|---|---|---|---|---|
| 1 | v1 | | | |
| 2 | | | | |
| 3 | assay | native-barcode-96-kit14:v0.5 | | |
| 4 | library_id | LibraryID | | |
| 5 | created_by | UserID | | |
| 6 | created_at | 2024-10-10T10:07:14+01:00 | | |
| 7 | sample_count | 48 | | |
| 8 | | | | |
| 9 | well_id | barcode | sample_type | sample_id |
| 10 | A1 | barcode01 | sample | sample1 |
| 11 | B1 | barcode02 | sample | sample2 |
| 12 | C1 | barcode03 | sample | sample3 |
| 13 | D1 | barcode04 | sample | sample4 |
| 14 | E1 | barcode05 | sample | sample5 |
| 15 | F1 | barcode06 | sample | sample6 |
| 16 | G1 | barcode07 | sample | sample7 |
| 17 | H1 | barcode08 | sample | sample8 |
| 18 | A2 | barcode09 | sample | sample9 |
| 19 | B2 | barcode10 | sample | sample10 |
| 20 | C2 | barcode11 | sample | sample11 |
| 21 | D2 | barcode12 | sample | sample12 |
| 22 | E2 | barcode13 | sample | sample13 |
| 23 | F2 | barcode14 | sample | sample14 |

## 4. ElysION run setup

**Materials**

For 1–4 samples: 270 fmol DNA in 20 µl per sample
For 5–48 samples: 110 fmol DNA in 20 µl per sample
Native Barcodes (NB01-NB96)
EDTA (EDTA)

| | |
|---|---|
| | AMPure XP Beads (AXP)<br>Long Fragment Buffer (LFB)<br>Elution Buffer (EB)<br>Native Adapter (NA) |
| **Consumables** | NEBNext® FFPE DNA Repair Mix (NEB, M6630)<br>NEBNext® Ultra™ II End Repair/dA-Tailing Module (NEB, E7546)<br>NEB Blunt/TA Ligase Master Mix (NEB, M0367)<br>NEBNext Quick Ligation Module (NEB, E6056)<br>Hard-Shell® 96-Well PCR Plates, low profile, thin wall, skirted, red/clear (Bio-Rad™, HSP9611)<br>Nunc™ 96 Well Polypropylene 2 ml DeepWell Plate (Thermo Scientific, 95040452)<br>ElysION Disposable Tips - 1000 µl (ELY-TIP1000)<br>ElysION Disposable Tips - 50 µl (ELY-TIP0050)<br>ElysION Disposable Tips Box Small (ELY-TBS01)<br>ElysION Disposable Tips Box Large (ELY-TBL01)<br>Reservoir Plate (Porvair, 390015)<br>Azenta PCR Plate Lid (Azenta, 4ti-0291)<br>Screw Cap Micro tube 2 ml, sterile (Sarstedt™, 72.694.006)<br>Screw Cap Micro tube 0.5 ml, sterile (Sarstedt™, 72.730.106)<br>ElysION Waste Bin (ELY-WB01)<br>Freshly prepared 80% ethanol in nuclease-free water<br>Nuclease-free water (e.g. Thermo Scientific, AM9937)<br>Bovine Serum Albumin (BSA) (50 mg/ml) (e.g Invitrogen™ UltraPure™ BSA 50 mg/ml, AM2616)<br>1.5 ml Eppendorf DNA LoBind tubes |
| **Equipment** | Microfuge<br>Vortex mixer<br>Microplate centrifuge, e.g. Fisherbrand™ Mini Plate Spinner Centrifuge (Fisher Scientific,11766427)<br>P1000 pipette and tips<br>P200 pipette and tips<br>P100 pipette and tips<br>P20 pipette and tips<br>P10 pipette and tips<br>P2 pipette and tips<br>Multichannel pipette and tips<br>Ice bucket with ice |

**The automated library preparation and sequencing method using the ElysION device can be followed using the on-screen display on the device.**

Please ensure you have all the correct hardware components, software packages and workflows installed to carry out this method.

For more information on the device and the processes, please refer to the ElysION User Guide.

1. **Thaw the sequencing kit components and third party reagents at room temperature, spin down briefly using a microfuge and mix by pipetting as indicated by the Table below:**

| Reagent | 1. Thaw at room temperature | 2. Briefly spin down | 3. Mix well by pipetting |
|---|---|---|---|
| NBD Barcode Plate (NB01-96) | Not frozen | ✓ | ✓ |
| Native Adapter (NA) | Not frozen | ✓ | ✓ |
| AMPure XP Beads (AXP) | ✓ | ✓ | Mix by pipetting or vortexing immediately before use |
| Elution Buffer (EB) | ✓ | ✓ | ✓ |
| EDTA | ✓ | ✓ | Mix by vortexing |
| Long Fragment Buffer (LFB) | ✓ | ✓ | Mix by vortexing |
| Flow Cell Flush (FCF) | ✓ | ✓ | ✓ |
| Flow Cell Tether (FCT) | ✓ | ✓ | ✓ |
| Sequencing Buffer (SB) | ✓ | ✓ | ✓ |
| Library Beads (LIB) | ✓ | ✓ | Mix by pipetting or vortexing immediately before use |
| Wash Mix (WMX) | ✓ | ✓ | ✓ |
| Wash Diluent (DIL) | ✓ | ✓ | Mix by vortexing |
| Storage Buffer (S) | ✓ | ✓ | Mix by vortexing |
| Bovine Serum Albumin (BSA) | ✓ | ✓ | ✓ |
| NEB Blunt/TA Ligase Master Mix (NEB, M0367) | ✓ | ✓ | ✓ |

| Reagent | 1. Thaw at room temperature | 2. Briefly spin down | 3. Mix well by pipetting |
|---|---|---|---|
| NEBNext® FFPE DNA Repair Mix | ✓ | ✓ | Mix by pipetting, do not vortex |
| NEBNext FFPE DNA Repair Buffer | ✓ | ✓ | Mix by vortexing |
| NEBNext® Ultra II End Prep Enzyme Mix | ✓ | ✓ | Mix by pipetting, do not vortex |
| NEBNext® Ultra II End Prep Reaction Buffer | ✓ | ✓ | Mix by vortexing |
| NEBNext Quick Ligation Reaction Buffer | ✓ | ✓ | ✓ |
| Quick T4 DNA Ligase | ✓ | ✓ | ✓ |

**Note:** Please ensure you prepare the NEB reagents in accordance with manufacturer's instructions.

**The wells of the barcoding plate are intended for single use only. Please ensure your barcode well is sealed before use, and do not reuse the barcode well once pierced/opened.**

## 2 Switch on the ElysION device and its computer following the user guide.

**The Background Services application must run continuously whilst using the ElysION, and should not be closed.**

## 3 Ensure the Background Services application is running.

If the application is not running, double click the **Background Services** desktop application on your ElysION device.

## 4 Open the ElysION UI application.

> **Ensure the ElysION deck is clear of any labware before performing automated checks.**
>
> Failure to ensure the deck is clear can lead to errors in checks or damage to the equipment.

## 5 Close the door to the robot and select "Initialise" on the display.

**Note:** The ElysION device will perform automated checks.

Allow the initialisation checks to complete before proceeding with the library preparation method.

## 6 Select "Run method" on the on-screen display.

## 7 Select the method “Native Barcoding Kit” on the on-screen display.

Click **Next** to proceed.

## 8 Insert a MinION and GridION Flow Cell into the MinION Mk1D device on the ElysION deck.

Instructions for inserting a flow cell onto the ElysION device can be found in the ElysION User Guide.

## 9 Select "Check flow cell" on the on-screen display.

Once flow cell check begins, click **Next** to proceed.

## 10 Select "Import" on the on-screen display to upload your sample sheet.

Once completed, click **Next** to proceed.

## 11 In a new 2 ml Sarstedt screw cap tube, prepare the End-Prep Master Mix as follows and pipette mix:

**Note:** The reagents, volumes required and total volume of the End-Prep Master Mix will vary depending on the input type and number of samples being processed. The volumes for your experiment will be displayed in the ElysION screen UI.

**For the gDNA method variant:**

| Reagent | Volume |
|---|---|
| FFPE Buffer | 29.2% |
| FFPE Enzyme | 16.6% |
| Ultra II Buffer | 29.2% |
| Ultra II Enzyme | 25% |
| **Total** | **X µl** |

**For the amplicon method variant:**

| Reagent | Volume |
|---|---|
| Ultra II Buffer | 70% |
| Ultra II Enzyme | 30% |
| **Total** | **X µl** |

## 12 Prepare the remaining reagents in accordance with the reagent preparation page on the on-screen display.

Once completed, click **Next** to proceed.

## 13 Once your flow cell check has successfully completed, set up your sequencing run conditions on the on-screen display:

1. Select the sequencing time limit, or to stop run when the flow cell pores are depleted.
2. Select a data target.
3. Select whether to perform:

- A flow cell flush to return the used flow cell to Oxford Nanopore Technologies.
- Or a flow cell wash using the Flow cell wash kit (EXP-WSH004) to reuse your flow cell.

**Note:** Reagent volumes and guidance for flow cell wash or flush will be given on the deck loading page. If you would like to perform a flow cell wash for re-use, we recommend thawing the wash reagents at the same time as the library preparation reagents.

For more information on the run conditions, please refer to the ElysION User Guide.

**Reagents and consumables requirements**

Please consider the following when preparing and loading your reagents and consumables into the ElysION device to mitigate risk of workflow failure and equipment damage:

- Ensure that the correct consumables are used with each reagent.
- Spin down all samples and reagents, making sure they do not contain bubbles.
- Ensure plates are flat in their deck position and sit within the barriers on the deck, and tubes are fully inserted into the rack.

### 14 Set up the ElysION deck according to the deck layout outlined in the UI of the on-screen display.

**Note:** The position of the tip boxes and reagents will change depending on the sample count and run setup. Please ensure you are following the instructions on the on-screen display correctly for your run.

### 15 Insert two empty waste bins into the disposable waste drawer below the deck.

### 16 Close the door of the ElysION device.

### 17 Click "Begin run" to start the automated library preparation and sequencing.

**Once the run is finished, unload labware, empty the bins, and discard or store in the freezer any unused reagents.**

# 5. Issues during the automated library preparation

### ElysION device troubleshooting

For commonly encountered issues, please refer to the ElysION User Guide.

For in-depth troubleshooting, please refer to the **Set-up and operating manual: Early access ElysION** provided with your ElysION device.

For additional customer support, contact the nanoporetech support channels.